

Week 4 – Initial System Configuration & Security Implementation

The purpose of Phase 4 is to ensure remote-only administration using secure authentication and access control from the Mint workstation. This entails:

- Verifying network connection between the two VMs
- Enabling SSH access
- Generating and using SSH key pairs for authentication
- Hardening the server by disabling password logins
- Ensuring that only authorised system can connect

These steps improve system security and follow best practices for administrating servers in real environments

Ping Test (Mint to Ubuntu server)

Before configuring SSH, I verified that the Mint workstation could communicate with the Ubuntu server using ICMP (ping)

The server was configured with the static IP: 192.168.56.10 and the workstation was configured with 192.168.56.11

Then, from the Mint terminal, I ran the command “ping 192.168.56.10”. Successful replies confirm that both systems were on the same Host-Only network and that routing was functioning correctly

```
katana@katana-VirtualBox:~$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=1.68 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=1.49 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=2.69 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=2.18 ms
64 bytes from 192.168.56.10: icmp_seq=5 ttl=64 time=3.02 ms
64 bytes from 192.168.56.10: icmp_seq=6 ttl=64 time=2.59 ms
64 bytes from 192.168.56.10: icmp_seq=7 ttl=64 time=2.90 ms
64 bytes from 192.168.56.10: icmp_seq=8 ttl=64 time=3.03 ms
64 bytes from 192.168.56.10: icmp_seq=9 ttl=64 time=2.11 ms
64 bytes from 192.168.56.10: icmp_seq=10 ttl=64 time=1.63 ms
64 bytes from 192.168.56.10: icmp_seq=11 ttl=64 time=3.38 ms
64 bytes from 192.168.56.10: icmp_seq=12 ttl=64 time=2.08 ms
64 bytes from 192.168.56.10: icmp_seq=13 ttl=64 time=1.74 ms
64 bytes from 192.168.56.10: icmp_seq=14 ttl=64 time=2.00 ms
64 bytes from 192.168.56.10: icmp_seq=15 ttl=64 time=2.31 ms
64 bytes from 192.168.56.10: icmp_seq=16 ttl=64 time=2.17 ms
64 bytes from 192.168.56.10: icmp_seq=17 ttl=64 time=2.48 ms
64 bytes from 192.168.56.10: icmp_seq=18 ttl=64 time=1.72 ms
64 bytes from 192.168.56.10: icmp_seq=19 ttl=64 time=1.55 ms
^C
--- 192.168.56.10 ping statistics ---
19 packets transmitted, 19 received, 0% packet loss, time 18059ms
rtt min/avg/max/mdev = 1.493/2.249/3.375/0.547 ms
katana@katana-VirtualBox:~$
```

SSH was installed with: “sudo apt install openssh-server”. Then, I verified that the SSH daemon was running: “sudo systemctl status ssh”.

```

katana@katana-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab
   Active: active (running) since Fri 2025-12-12 19:15:26 GMT; 3s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 5410 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 5412 (sshd)
       Tasks: 1 (limit: 2255)
      Memory: 1.2M (peak: 1.5M)
         CPU: 20ms
    CGroup: /system.slice/ssh.service
            └─5412 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 12 19:15:26 katana-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Sec
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on 0.0.0.0 port 2
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on :: port 22.
Dec 12 19:15:26 katana-VirtualBox systemd[1]: Started ssh.service - OpenBSD Sec
(lines 1-18/18 (END))...skipping...
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-12-12 19:15:26 GMT; 3s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 5410 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 5412 (sshd)
       Tasks: 1 (limit: 2255)
      Memory: 1.2M (peak: 1.5M)
         CPU: 20ms
    CGroup: /system.slice/ssh.service
            └─5412 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 12 19:15:26 katana-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on 0.0.0.0 port 22.
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on :: port 22.
Dec 12 19:15:26 katana-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
~
~
~

```

Mint was configured with a static host-only IP. Then to connect remotely, I used: “[sudo vboxuser@192.168.56.10](#)”.

```

katana@katana-VirtualBox:~$ ssh vboxuser@192.168.56.10
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:WlrSdFea0iw2ItTF4BgGIn6UNuYXKB/eOB45gcPgwpU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ED25519) to the list of known hosts.
vboxuser@192.168.56.10's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-49-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

487 updates can be applied immediately.
235 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
vboxuser@Ubuntu:~$

```

On the Mint workstation, I generated a 4096-bit RSA key pair: “`ssh-keygen -t rsa -b 4096 -C “student@workstation”`”

```

katana@katana-VirtualBox:~$ ssh-keygen -t rsa -b 4096 -C student@workstation
Generating public/private rsa key pair.

Enter file in which to save the key (/home/katana/.ssh/id_rsa): /home/katana/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/katana/.ssh/id_rsa
Your public key has been saved in /home/katana/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:GtLesahncalcpZ5GjVv9huX49Ez8++YQ9ZhlY5AGoI student@workstation
The key's randomart image is:
+---[RSA 4096]-----+
|
|  E . . .
|  . = . O +
|  . * * = . . O
|  . . * S . . O .
|  . . * + +
|  . = O . . = B
|  . * . + O * O
|  . O . . . O + +
+---[SHA256]-----+
katana@katana-VirtualBox:~$

```

Then I copied the public key in to the server:

```

katana@katana-VirtualBox:~$ ssh-copy-id vboxuser@192.168.56.10
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/katana/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vboxuser@192.168.56.10's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vboxuser@192.168.56.10'"
and check to make sure that only the key(s) you wanted were added.

katana@katana-VirtualBox:~$

```

To improve server security, I disabled password authentication to enforce key-only access:

This was achieved by first editing: “sudo nano /etc/ssh/sshd_config”. Then setting password authentication to no and public key authentication to yes. After that I restarted the SSH.

Lastly, I enabled the UFW firewall:

```

katana@katana-VirtualBox:~$ sudo apt install ufw
[sudo] password for katana:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 to upgrade, 0 to newly install, 0 to remove and 31 not to upgrade.
katana@katana-VirtualBox:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
katana@katana-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
katana@katana-VirtualBox:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

katana@katana-VirtualBox:~$

```